Velli — Developer Build Task

75 minutes · any AI tool and the internet allowed · ask questions any time

Velli gives factory workers advances against wages they have already earned. The workers come under labour contractors, and the advance is recovered from their salary on payday. Build the tool our operations team will use to answer one question: **how much can this worker draw today?**

Your files

- **attendance_register.csv** — one row per worker: emp_code, worker_name, contractor, grade, per_day_rate, status, then one column per date. Marks: **P** = present, **A** = absent, **WO** = weekly off, blank = no entry.
- **draws.csv** — advances already paid out: draw_date (dd/mm/yyyy), worker_name, contractor, amount_rs, paid_to_name (whose UPI account got the money — sometimes a family member), upi_ref.

The rules

1. **Wage cycle** runs from the 10th of one month to the 9th of the next. Payday is the 10th.
2. **Earned so far** = per-day rate × paid days, from the start of the cycle up to and including the chosen date. Present days are paid. Absent and blank days are not.
3. **PF + ESI** (Provident Fund + Employees' State Insurance) = ₹2,000 per cycle, prorated: ₹2,000 × (calendar days from cycle start to the chosen date, inclusive) ÷ (total days in the cycle).
4. **Net earned** = earned so far − prorated PF + ESI.
5. **Eligible to draw** = 50% of net earned − amounts already drawn this cycle (amounts only, not fees). Maximum ₹5,000 per draw. Round down to the nearest ₹100. Never below zero.
6. **Inactive** workers can't draw.
7. **Fee** = 2% of the amount drawn. The worker receives the full amount; the fee is collected with the repayment on payday.
8. **Ignore anything dated after the chosen date** — attendance and draws.

What must work at 75 minutes

Input a worker and a date (and optionally a requested amount). Output: the cycle dates, paid days, earned so far, PF + ESI, amount already drawn, eligible amount, and — for a requested amount — approved or rejected, the fee and the total repayable. Web page, script or anything else, your choice.

Good to know

- This is like real factory data: it is messy. Handle it sensibly and tell us what you found.
- If a rule is unclear, ask us — the way you would at work. Don't guess silently.
- At the end we will type our own test cases into your tool. Correct numbers matter more than looks.
- Finished early? Add (a) a short WhatsApp message in Tamil and English telling the worker their limit, then (b) a table of every worker's limit on a chosen date.